

Informe Café → Azul — copia fiel (sin suavizar)

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PDF: INFORME_COPIA_AZUL_FIEL_V20b.pdf

Repasada general (evidencia = 16 parejas AAC / 32 M4A):

- Separación nivel vs timbre: gain ≈ -12 dB (consenso V12/V15/V17/V4.1)
- EQ operativa: V20.0 operativa · air v20_hard_10k
- pipeline_version: V20.0-operative
- source_variant: v19_presence_robust+v20_hard_10k
- air_policy: v20_hard_10k
- Gain operativa: -12.012 dB
- Hold-out crítico 0.5–8 kHz: 4.100 dB
- Sin suavizado regional ni contracción EQ×reliability
- Aire 10–18 kHz: taper V20 (eq_copy $\neq$ V19 cruda encima de 10 kHz)

Qué hacemos mejor que V4.1: DPSS, F0 refine, matching acústico, offsets estimados, repro bit-idéntica, métrica render Café+EQ vs Azul.

Qué adoptamos de V4.1 para fidelidad: pesos SNR/ciclos/AAC, pair-first, confianza de alineación, repetibilidad de fase, neutralidad energética.

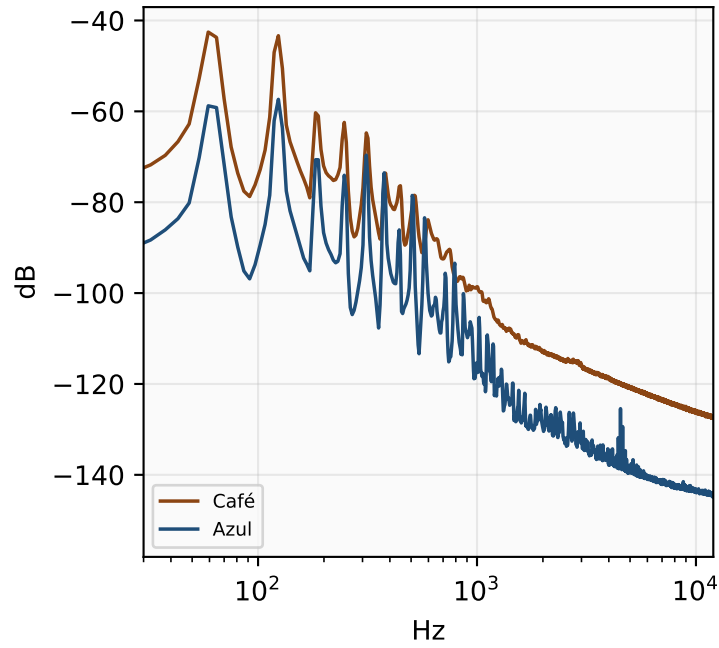
Qué NO adoptamos como EQ principal: smooth held-out, EQ×fiabilidad, multiplicadores físicos fijos 0.62/0.55/0.85.

Contenido:

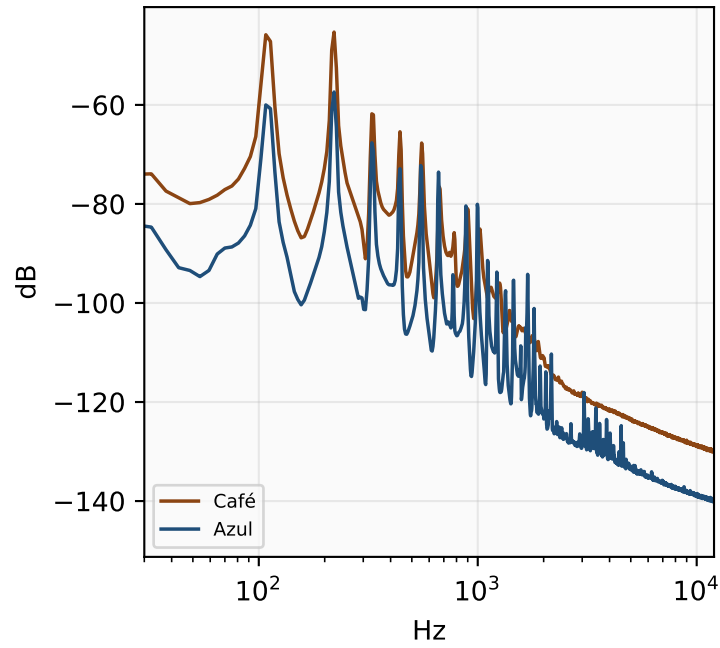
- 1) Espectros Café vs Azul (pares representativos)

1. Espectros Café vs Azul (Welch, frames activos)

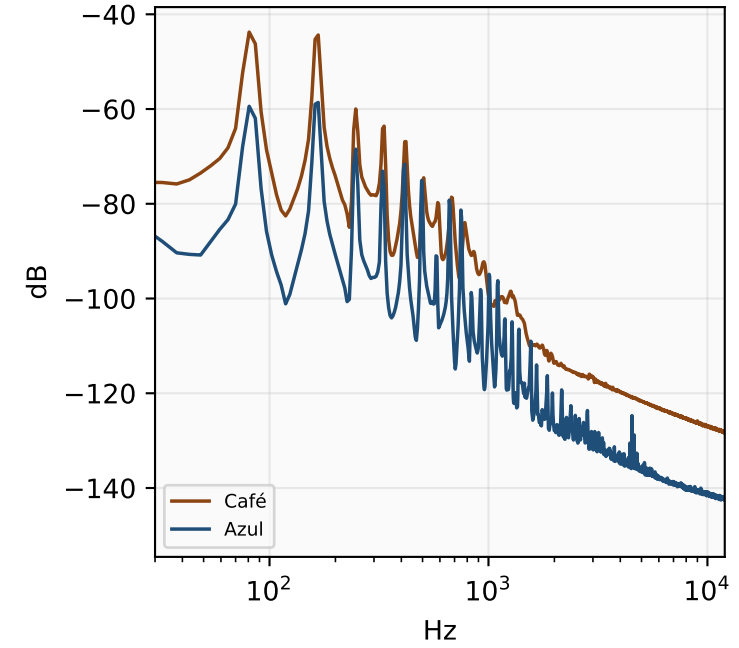
B₁₂



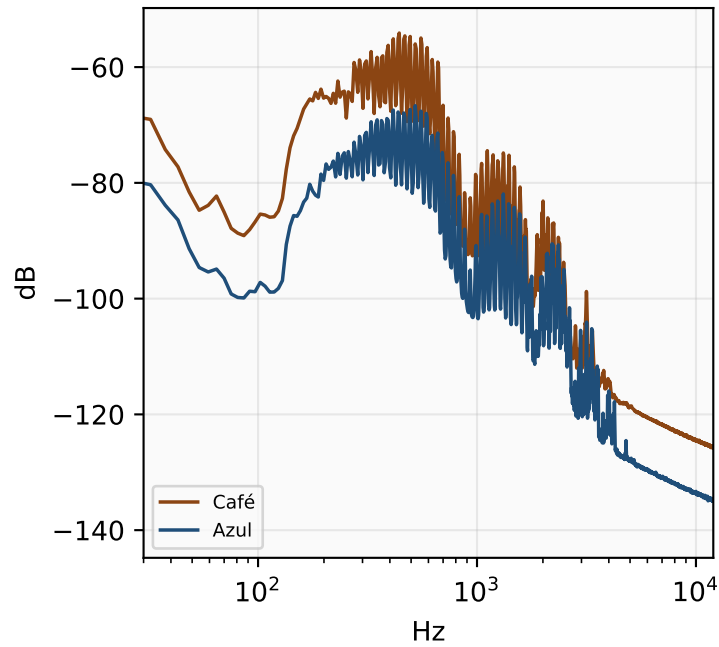
A₁₂



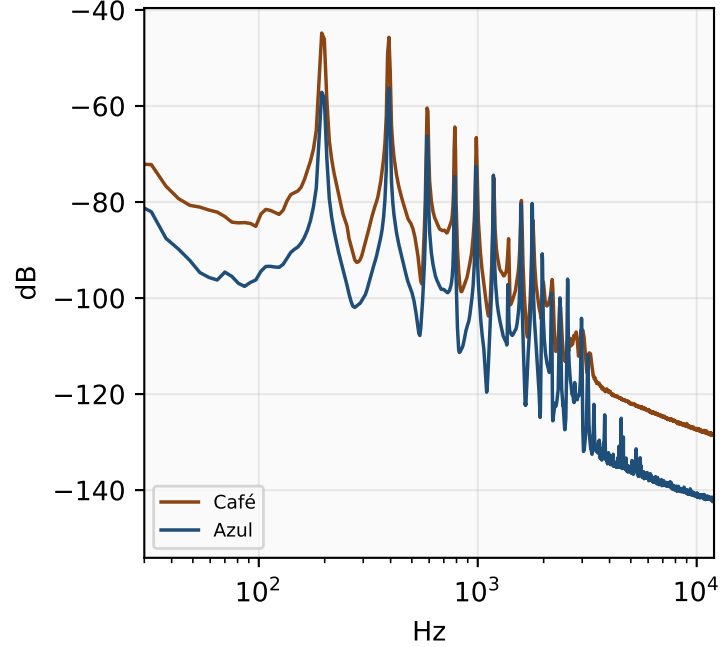
E₁₂



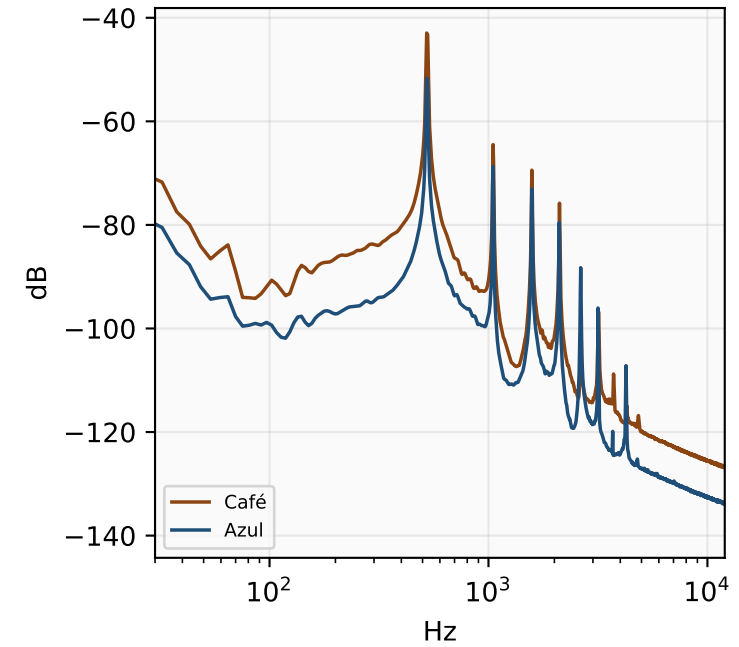
C_{chromatic}



G₁₂

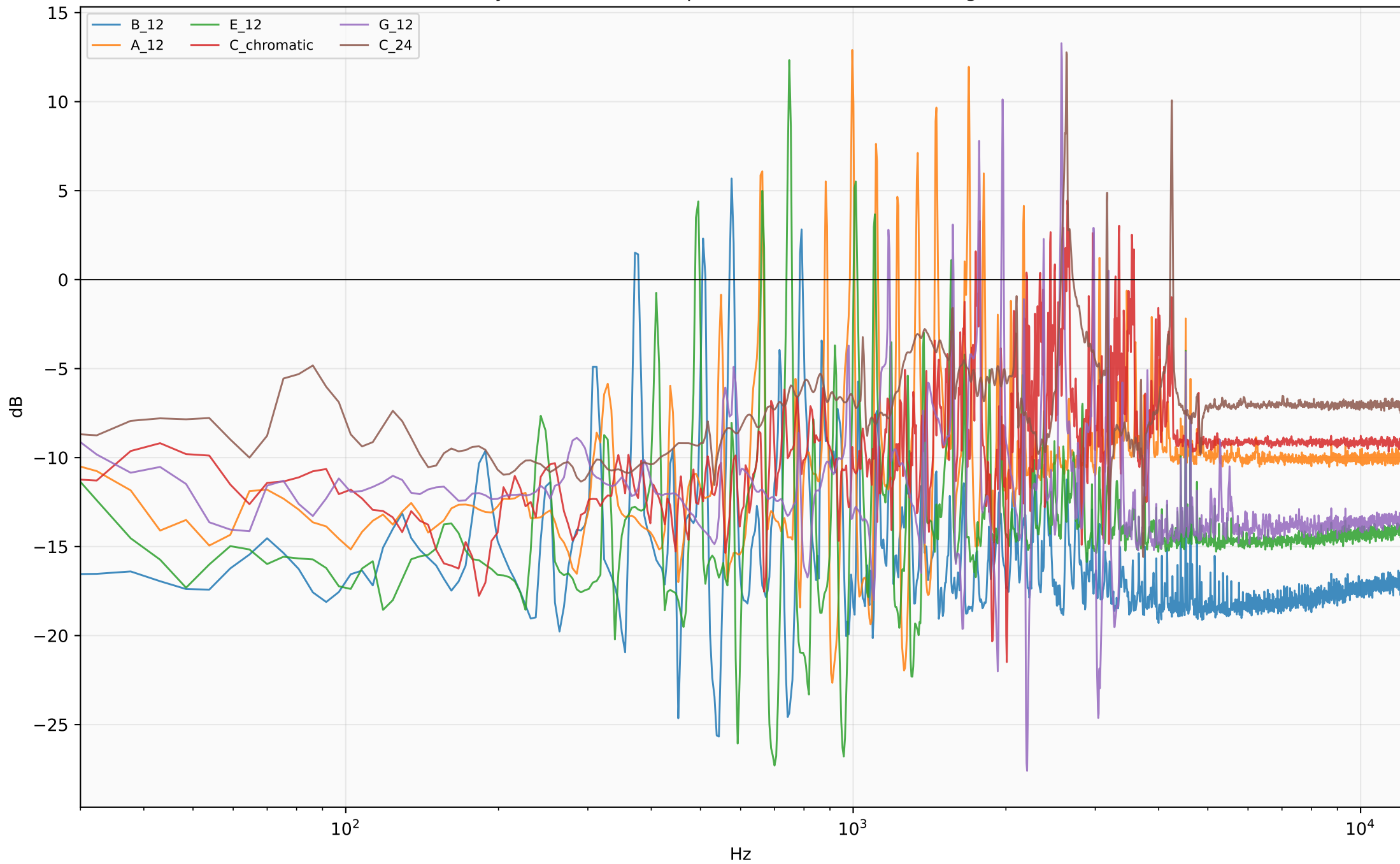


C₂₄



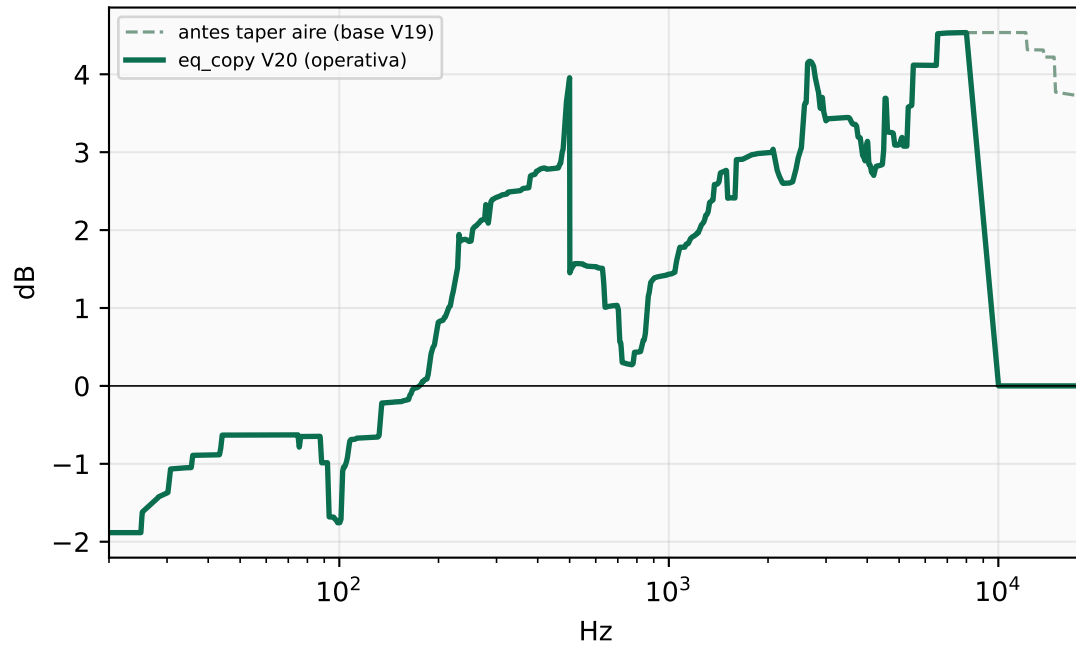
1b. Diferencia espectral Azul – Café (por pareja, sin EQ)

Incluye nivel + timbre (por eso ~ -12 dB de offset global)



2. EQ operativa fiel — V20.0 operativa · air v20_hard_10k (gain -12.01 dB)

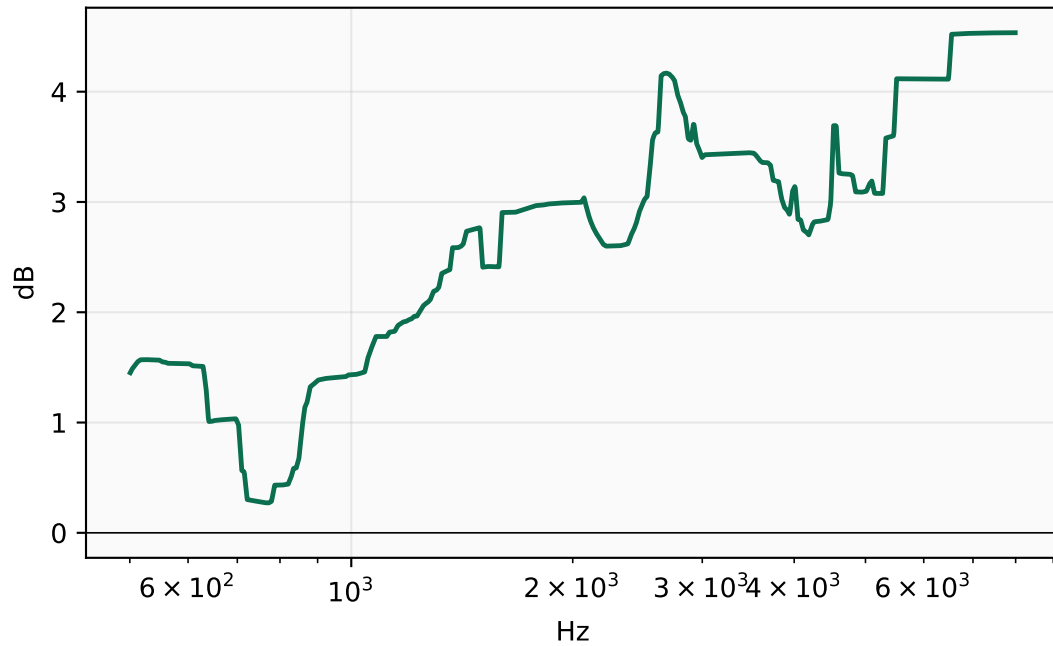
Timbre (sin gain)



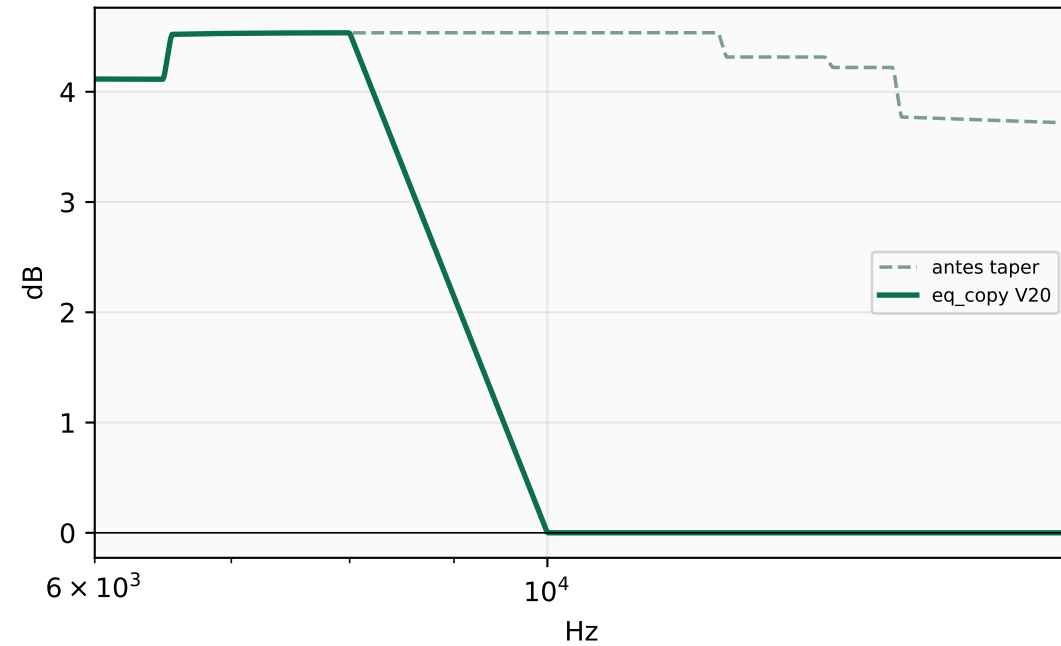
Total a aplicar al Café



Zoom 0.5–8 kHz (copia crítica)

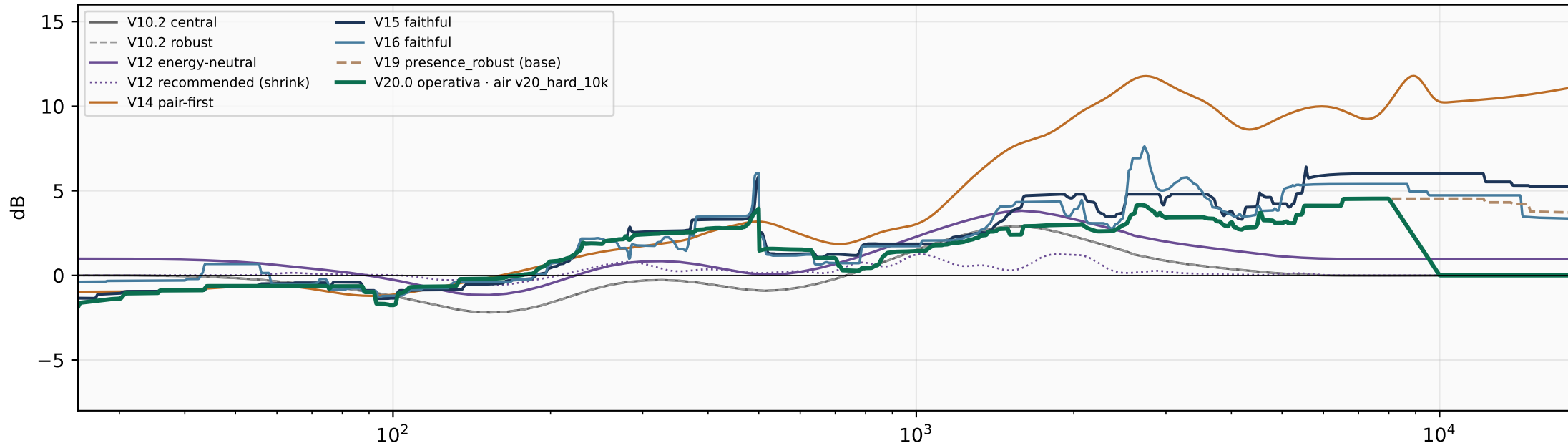


Zoom aire 6–18 kHz (v20_hard_10k)

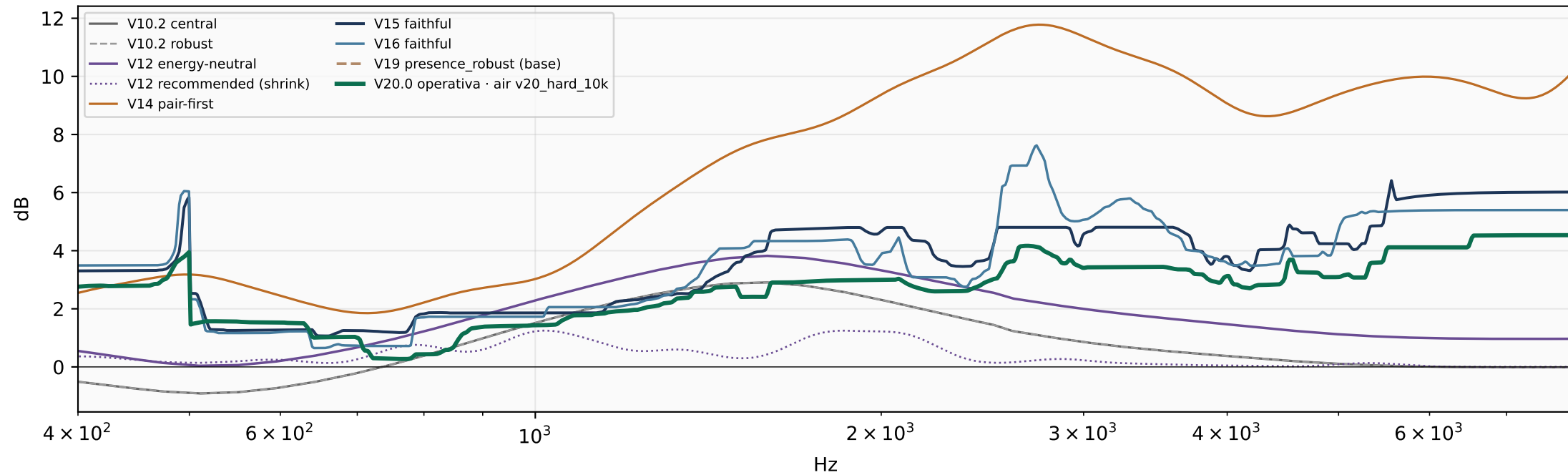


3. Todas las curvas de transferencia Café → Azul

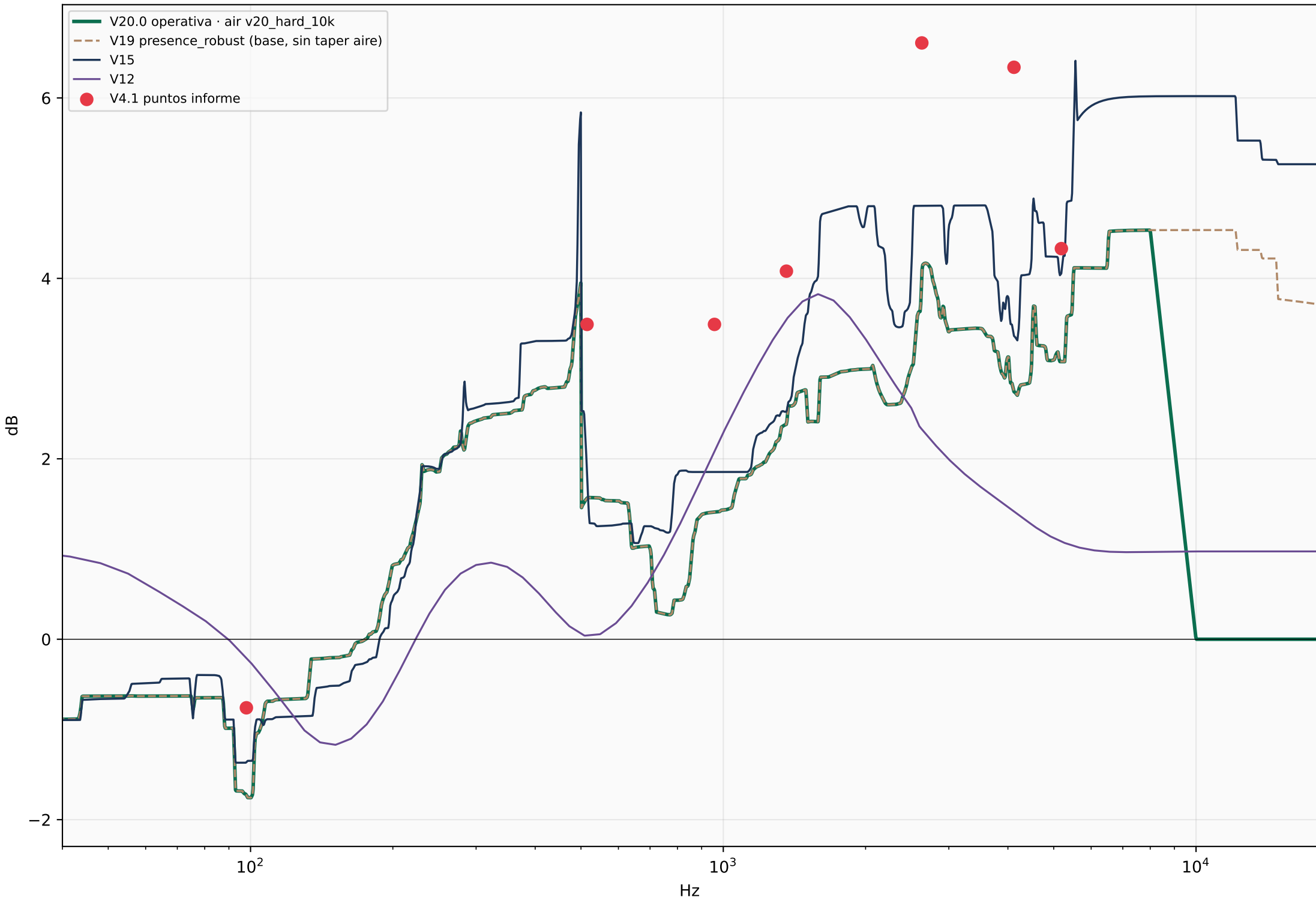
Comparación completa (timbre, sin gain global)



Zoom presencia / medios (0.4–8 kHz)

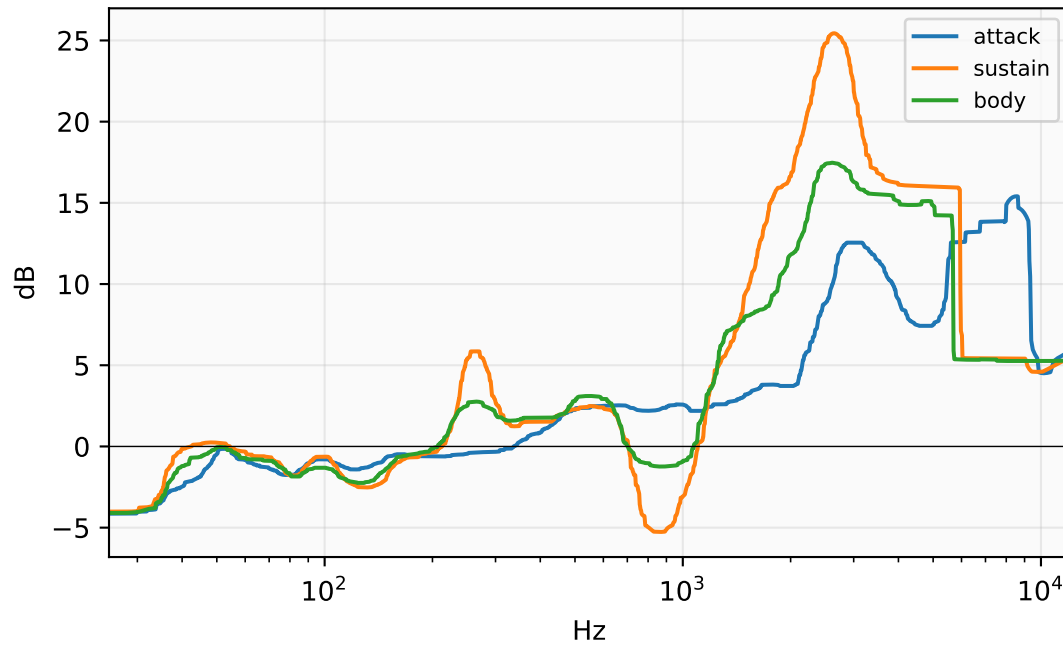


3b. Operativa V20 vs V19 base / puntos V4.1



4. Curvas por fase temporal (V13)

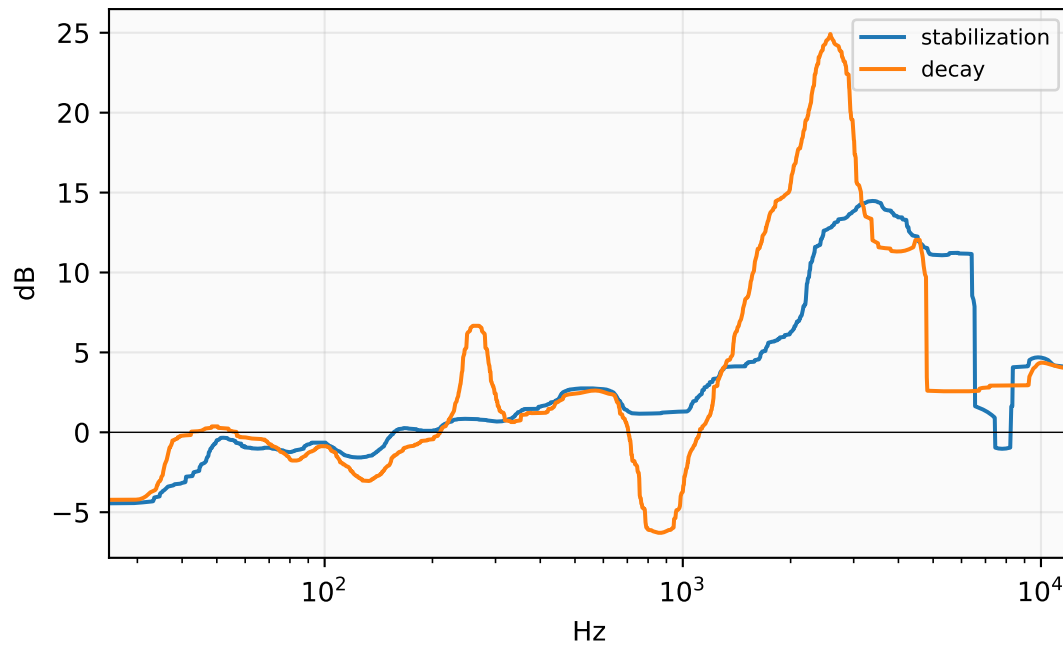
Ataque / sustain / cuerpo



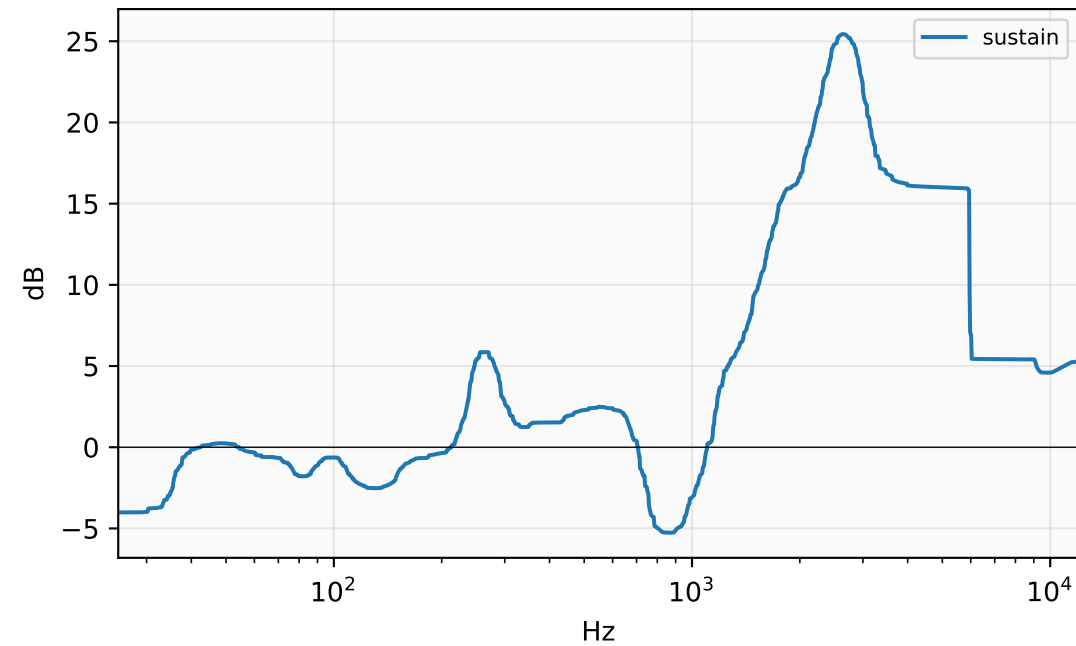
Ataque – cuerpo



Estabilización / decay

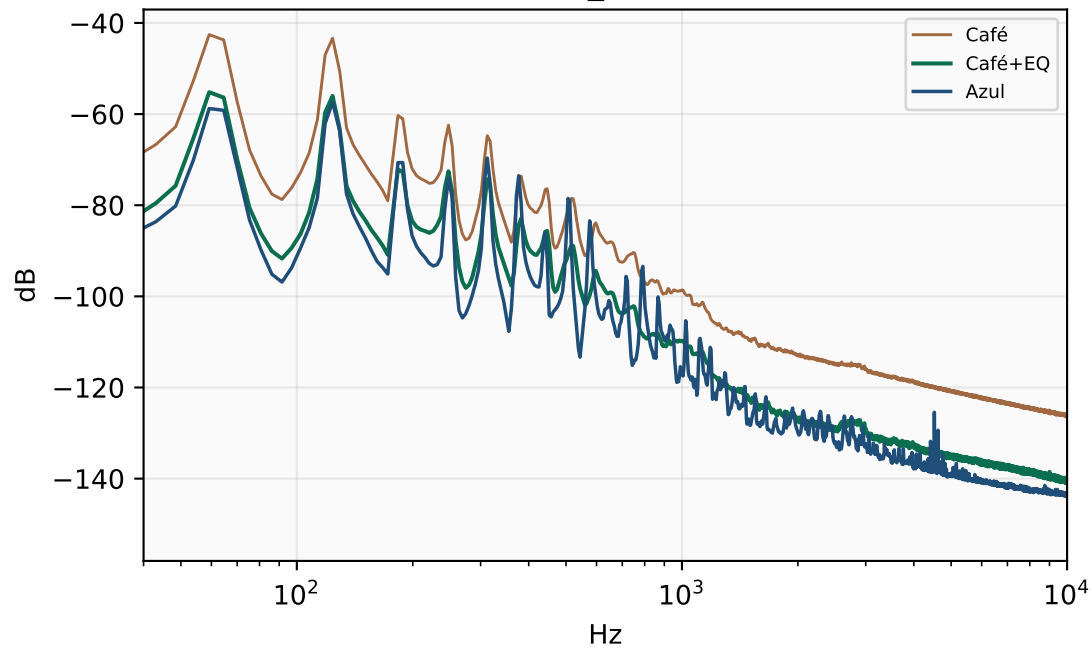


Sustain (detalle)

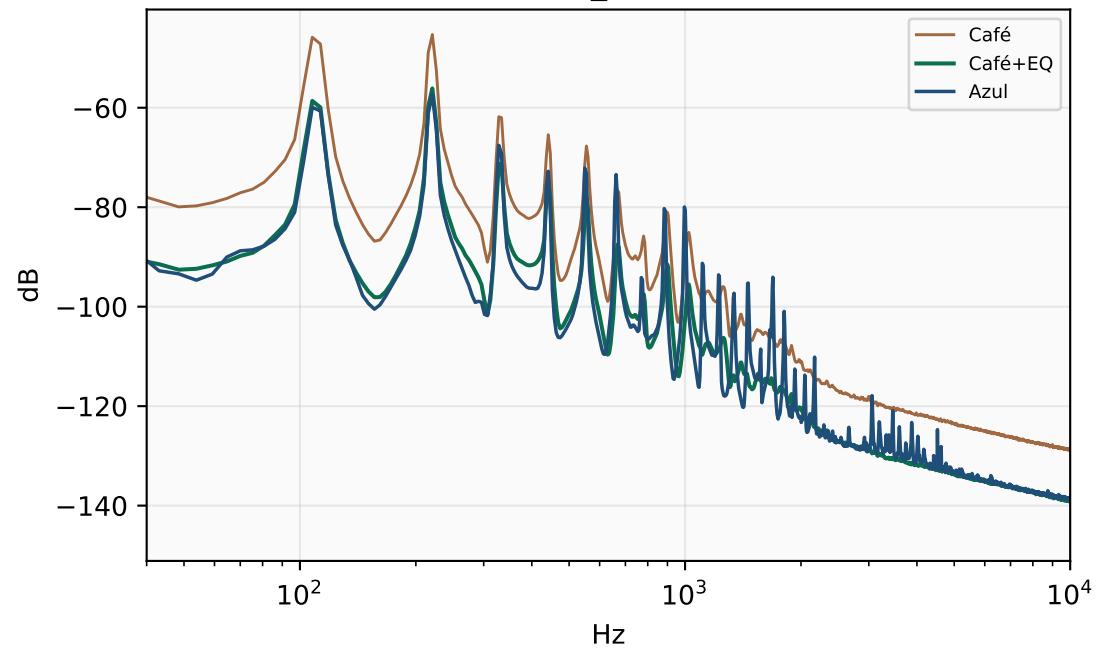


5. Fidelidad de copia: Café → (+EQ+gain -12.01 dB) vs Azul

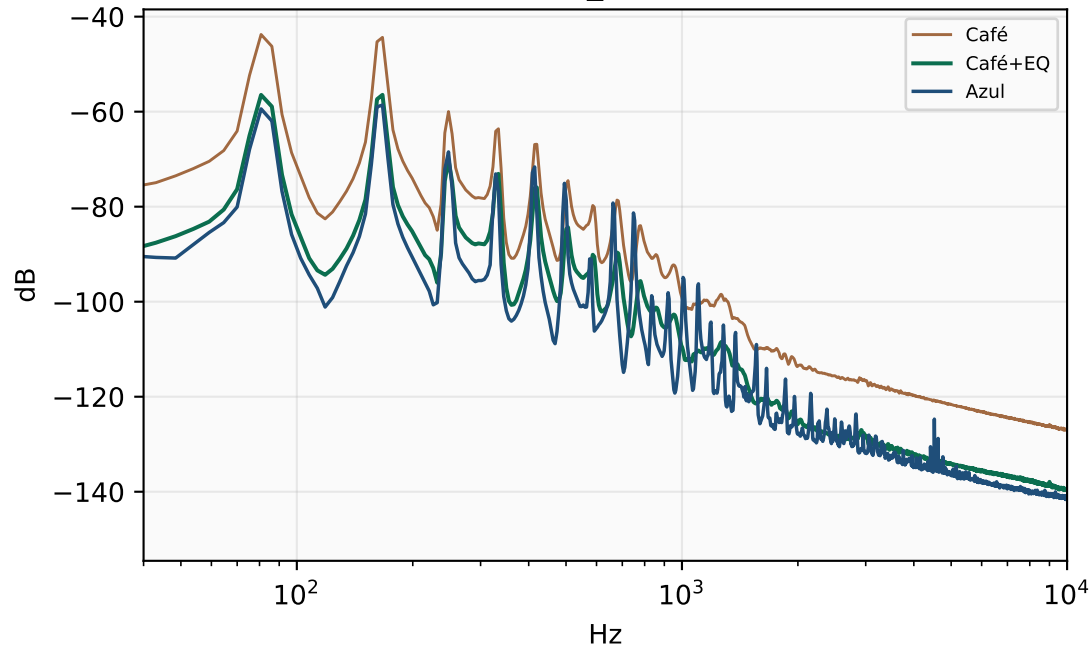
B_12



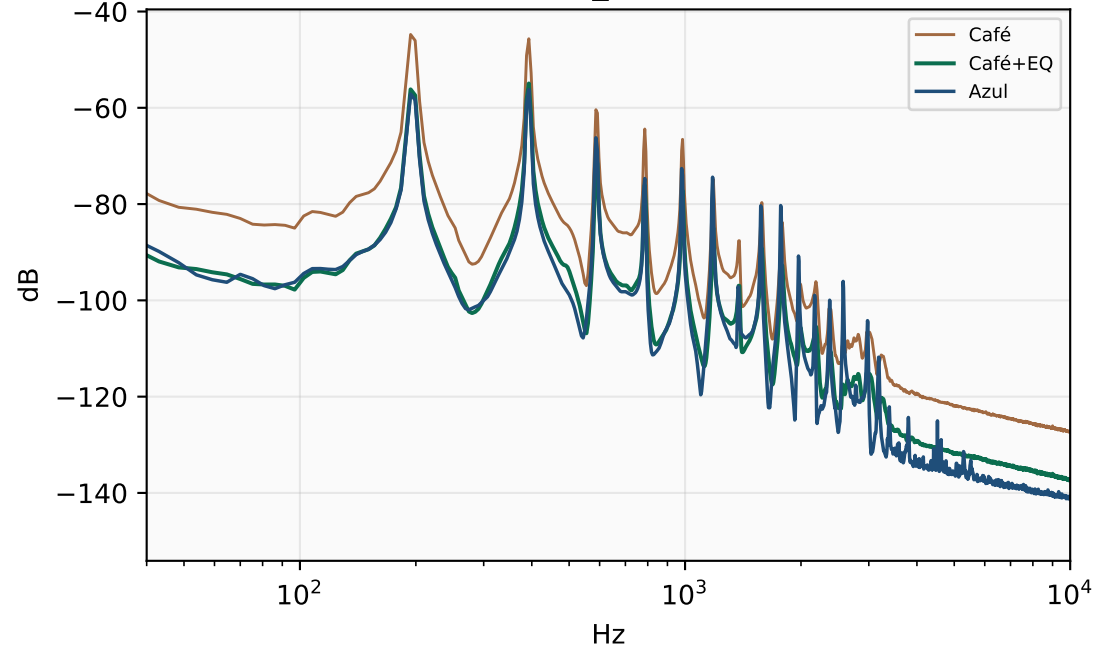
A_12



E_12



G_12

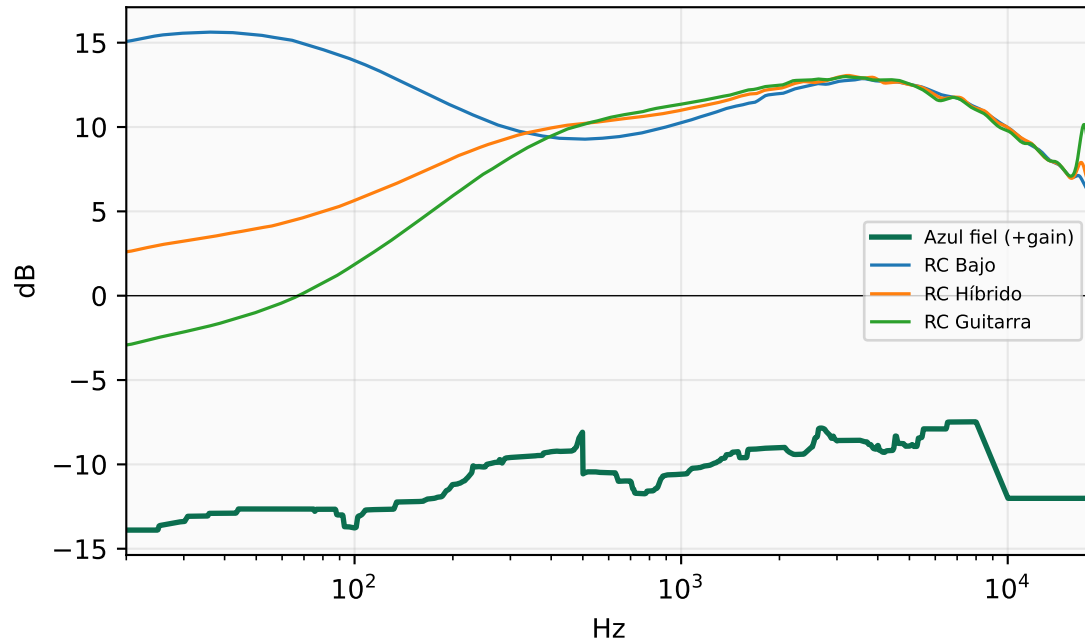


5b. Ranking aire V20 (operativa actual)

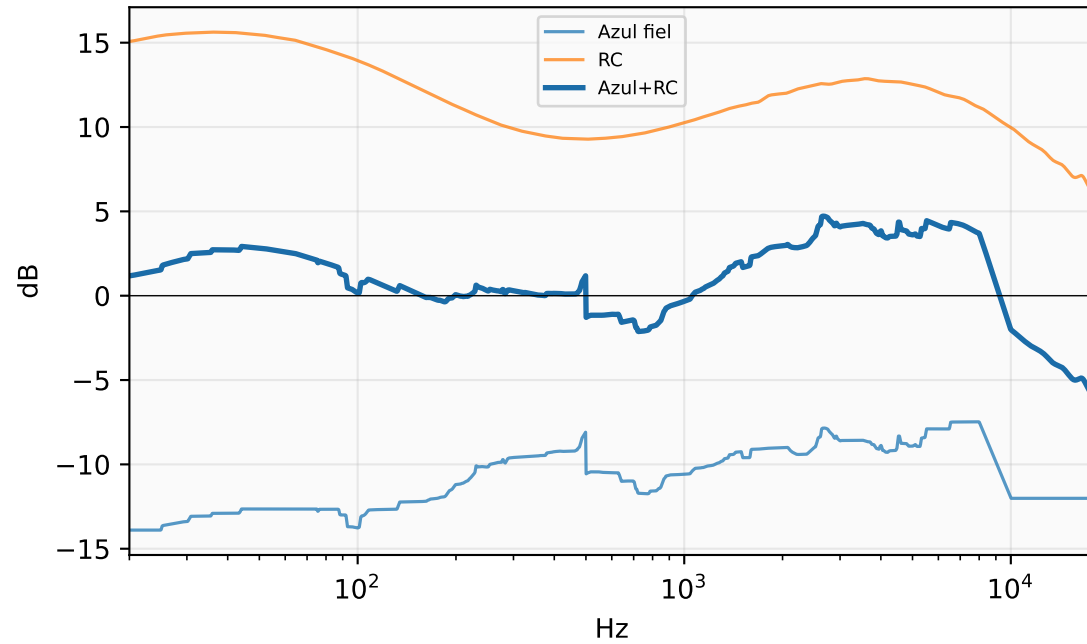
variant	holdout_critical_rmse_db	bias_2k4k_db	bias_8k12k_db	eq_at_10k_db	eq_at_15k_db	eq_at_18k_db	gain_db
v20_no_taper	4.100	0.056	0.555	4.536	3.769	3.715	-12.012
v20_aac_prior	4.100	0.056	0.409	4.087	1.264	0.297	-12.012
v20_soft_8_15	4.100	0.056	0.297	2.926	0.003	0.000	-12.012
v20_hard_10k	4.100	0.056	0.091	0.003	0.000	0.000	-12.012
v20_aac_then_soft	4.100	0.056	0.363	4.087	0.299	0.000	-12.012
v12_recommended_ref	5.313	-1.955	-0.862	0.000	0.000	0.000	-12.038

6. RC + EQ fiel Azul (gain -12.01 dB)

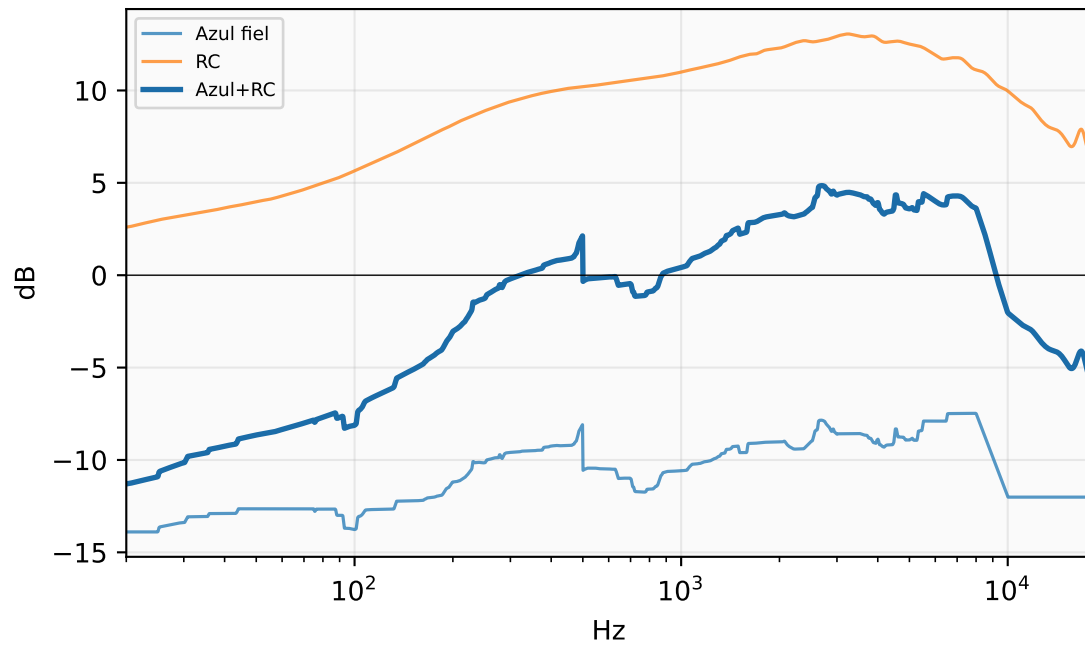
Componentes



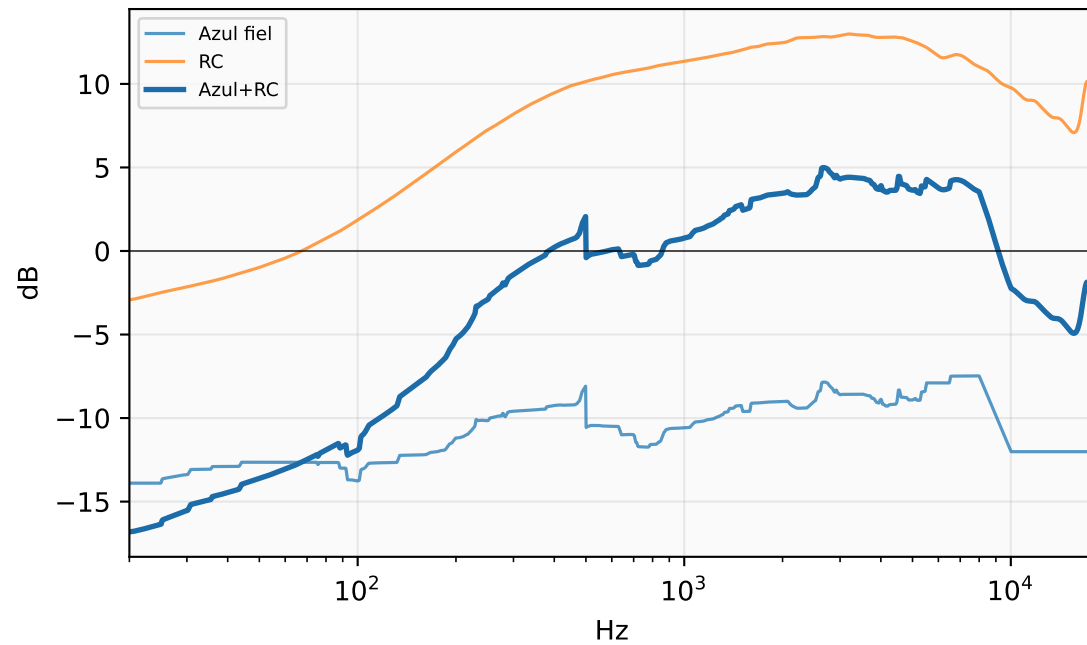
Composición Bajo



Composición Híbrido



Composición Guitarra



7. Cierre — estado de la copia Azul

EQ operativa: V20.0 operativa · air v20_hard_10k

pipeline_version: V20.0-operative

source_variant: v19_presence_robust+v20_hard_10k

air_policy: v20_hard_10k

Gain: -12.012 dB (reducir el Café)

Smoothing: none

Puntos de la curva operativa (eq_copy_db = V20 con taper aire):

98 Hz → -1.73 dB

515 Hz → +1.56 dB

958 Hz → +1.41 dB

1360 Hz → +2.38 dB

2630 Hz → +4.03 dB

4120 Hz → +2.74 dB

5190 Hz → +3.08 dB

8000 Hz → +4.53 dB

10000 Hz → +0.00 dB

15000 Hz → +0.00 dB

18000 Hz → +0.00 dB

Archivos clave:

· CURVA_COPIA_OPERATIVA.csv / GAIN_COPIA_OPERATIVA.csv

· RESUMEN_V20.json / FIDELIDAD_RANKING_AIRE_V20.csv

· FIDELIDAD_SIN_SUAVIZAR.md / MAPA_METODOLOGICO_V41_ULTRAPROFUNDO.md

· Audios: renders/FIDELIDAD_V20/...